

States, for example, pre-flight checks on baggage were finally introduced in 1972. Armed air marshals were placed on selected flights. A rash of D.B. Cooper copycats parachuting to the ground with ransom money was stopped by simple design changes that made it impossible to jump out of an airliner with a parachute. In 1974 the ICAO, the governing body of international aviation, agreed minimum security standards that all airlines had to meet, including checks on baggage loaded in the hold and on passengers and their hand luggage, plus measures to prevent unauthorized access to aircraft.

Israeli and Western governments took an increasingly hard line over hijackings, leading to spectacular incidents that were not necessarily reassuring for nervous passengers. In July 1976, the Israelis flew paratroops into Entebbe airport, Uganda, where Palestinian and German Baader-Meinhof terrorists were holding Jewish passengers from an Air France Airbus hostage, with the complicity of Idi Amin's Ugandan government. Thirteen terrorists were killed in the assault; three hostages were also killed, but the rest were taken back to Israel and safety. In October 1977, another combined Palestinian-Baader-Meinhof hijacking ended with an assault on the hijacked aircraft by German and British special forces at Mogadishu, Somalia, in which three terrorists were shot dead and the hostages were freed.

Tightening security and a tougher response definitely made it harder for hijackers and terrorists to

“The pictures of airplanes flying into buildings, fires burning, huge structures collapsing, have filled us with disbelief, terrible sadness, and a quiet, unyielding anger...”

PRESIDENT GEORGE W. BUSH

IN HIS ADDRESS TO THE NATION, SEPTEMBER 11, 2001

operate, but too many loopholes existed in the system for them to be stopped completely. In June 1985, for example, TWA Flight 847 was hijacked by Islamic Hezbollah terrorists from Lebanon, who murdered one American on board and held 39 others hostage for 17 days, while Sikh extremists blew up an Air India jumbo jet over the Atlantic, killing all 329 people on board.

Stopping explosive devices from being hidden in baggage was a challenge that seemed to defy solution. Just before Christmas 1988, a Pan Am 747 flying from Frankfurt to New York via Heathrow exploded at 31,000ft (9,400m) over the small Scottish town of Lockerbie, killing 259 people in the air and 11 on the ground. The cause was a bomb in the baggage hold.

September 11, 2001

To a few individuals, this sorry catalog of misery and disaster was an inspiration. Islamic extremists such as Osama bin Laden, obsessed by the goal of humiliating the United States with spectacular acts of terror, saw airliners as both a vulnerable and a symbolic target, just as the Palestinians had in 1970. What represented Western civilization more completely than the airplane, so often a key element in Western dominance? In the 1990s, the American public was informed that the FBI had uncovered an extraordinary plot by Islamic extremists to hijack 12 airliners simultaneously and destroy them in an airborne holocaust. It sounded the stuff of fantasy, but was not.

On September 11, 2001, terror in the air reached its apogee. As had happened before, terrorists exploited carefully researched weaknesses in the system, targeting American domestic flights, on which security was relatively lax, and relying on blades rather than guns – much easier to pass through security checks. They also exploited the received wisdom that governs response to hijacking – the well-drilled pattern of non-resistance and cooperation with the hijackers in which aircrews are trained, and the fearful but calm compliance of passengers. No one had thought of hijackers having trained as pilots, turning airliners into guided missiles. Of the four besieged airplanes, one was slammed into each of the Twin Towers of New York City's World Trade Center and one struck the Pentagon building in Washington, D.C. The fourth aircraft fatally crashed in a field in rural Pennsylvania when the heroic passengers and crew overpowered their assailants.

The events of September 11, 2001, had only a short-lived impact on the public's confidence in the safety of air travel. Tougher security measures at airports kept the terrorist threat in check, even if it could not be completely eliminated – as in the Russian-operated Airbus 321 blown up by an onboard bomb over Egypt in October 2015. Each major fatal accident presented a challenge to air-safety authorities, called upon to close a loophole or tighten a regulation. When a deranged copilot deliberately crashed an airliner in the Alps in March 2015, personality checks on pilots and rules governing two-person crews came under scrutiny. When an airliner was shot down by a missile over Ukraine in July 2014, questions were raised about routing of flights over war zones. With around 3.5 billion passengers flying worldwide every year by 2015, a death toll in air accidents averaging around 1,000 a year meant flight had probably become as safe as it could ever get.

TRADE CENTER TERROR

As one tower of the World Trade Center burns, a second airliner heads for the tower's twin. Using airliners as human-guided missiles, with their fuel to wreak destruction in place of a warhead, the terrorists were able to inflict vast damage while armed with nothing more than sharp blades.

SECURITY MEASURES

Airport security has required continual upgrading to cope with an evolving terrorist threat, causing substantial inconvenience to travelers. Above are items confiscated from passengers at Beijing's main airport, and, right, a scanner in action at Orlando International, Florida.

